

Early mechanical devices

Once people had settled down in permanent towns and villages, they began to invent and build devices to help them with their everyday tasks, such as processing food. Many of the most important early devices were for producing materials that could be made into clothes. People mostly used them in their own homes or in small workshops.

First handloom

- **What?** Weaving loom
- **Who?** Ancient Egyptians
- **Where and when?** Egypt, c. 5000 BCE

Cloth is made by intertwining lengths of thread placed at right angles to each other. The handloom was invented to make this process easier and quicker. Early looms were simple frames that held vertical threads (called the warp) firmly in position, while horizontal threads (called the weft) were woven through them.



Model of a handloom used by early European people, 800–600 BCE

First drawloom

- **What?** Pattern weaving loom
- **Who?** Ancient Chinese
- **Where and when?** China, c. 400 BCE

The drawloom, which allowed greater control over the threads than the handloom, was invented to weave highly patterned cloth, mainly silk. The key part was an arrangement called the “figure harness” that lifted individual warp threads. Drawlooms were large devices, often 13ft (4m) long, and required two people to operate them.

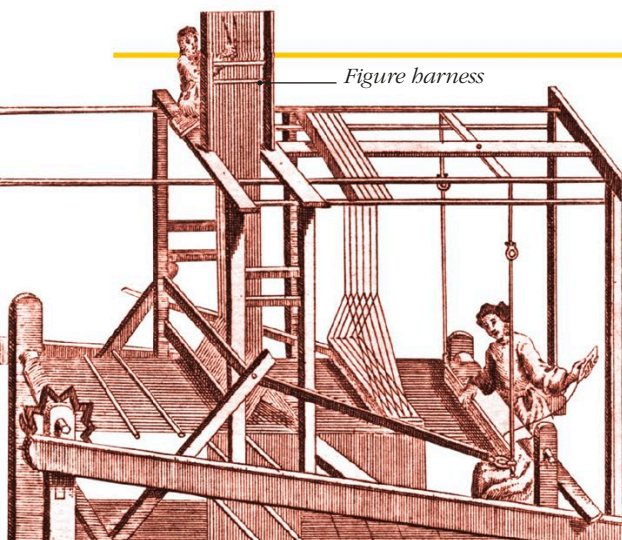
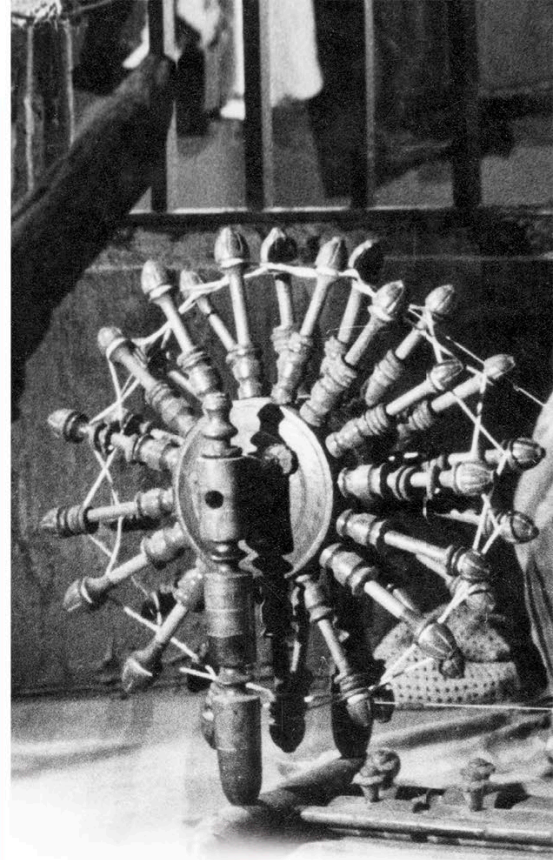


Figure harness



A woman in India turns a spinning wheel to produce thread

Rotary quern

- **What?** Hand mill for grinding grains
- **Who?** Unknown
- **Where and when?** Southern Europe, c. 600 BCE

Quern-stones grind cereal grains such as wheat into flour. The rotary quern consists of two circular stones, one placed on top of the other. The bottom one, called the “quern,” does not move. The top part, or “handstone,” has a handle attached so it can be rotated. As the quern turns, it crushes the grain that is fed in through a hole in the center.

Rotary querns are still used in some parts of the world



Wooden handle to turn handstone

Hole for grain